$\frac{q}{\varepsilon_0} = \oiint \vec{E} d\vec{S} = \oiint \frac{\vec{j} dS}{\sigma} = \frac{I}{\sigma}$

$\frac{dq}{dt}\frac{1}{\sigma} = -\frac{1}{\varepsilon_0} q$

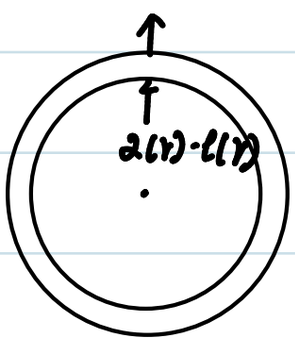
$\Rightarrow -\frac{dq}{dt} = -\frac{1}{\varepsilon_0 \sigma} q$

$\Rightarrow q = q_0 e^{-\frac{1}{\varepsilon_0 \sigma} t}$

$\frac{dq}{dt} = \frac{d\sigma}{dt} \cdot 2\pi r \cdot dr$

$= -q v r \cdot dr \cdot \frac{r^2 - 2h^2}{(h^2+r^2)^{5/2}}$

$2(r+dr) \cdot l(r+dr)$



$2(r+dr)l(r+dr) - 2(r)l(r) = \frac{dq}{dt}$

$\Rightarrow d(2 \cdot 2\pi r) = \frac{qv}{2} \cdot \frac{r^2 - 2h^2}{(h^2+r^2)^{5/2}} \cdot d(r^2)$

$\Rightarrow 2 \cdot 2\pi r = qv \cdot \frac{r^2}{(h^2+r^2)^{3/2}}$

$\Rightarrow 2 = \frac{qv}{2\pi} \frac{r}{(h^2+r^2)^{3/2}}$

(3). $M = \int dm = \int B I dr \cdot r$

$= \int \frac{A}{r^2} \cdot 2 \cdot 2\pi r \cdot r \cdot dr$

$= \int 2\pi \frac{Aqv}{2\pi} \frac{r}{(h^2+r^2)^{3/2}} dr$

$= \frac{qvA}{2} \int_0^\infty \frac{d(r^2)}{(h^2+r^2)^{3/2}}$

$= \frac{qvA}{h}$

So 类:

1 (1) 椭圆

能

角:

$\Rightarrow \frac{1}{2} m v_1^2 - \frac{GMm}{r_{ES}} = -\frac{GMm}{r_{EST} r_{MS}}$

$\Rightarrow v_1 = ?$

地球速度.

$\Rightarrow v_E = \sqrt{\frac{GM}{r_{ES}}}$

地球附近. v_0' 是离地∞处的速度.

$\frac{1}{2} m v_0^2 - \frac{GMm}{r_E} = \frac{1}{2} m v_0'$ (太阳引力忽略).

的来: $V_E - V_0' = V_1$ (以太阳系为系).

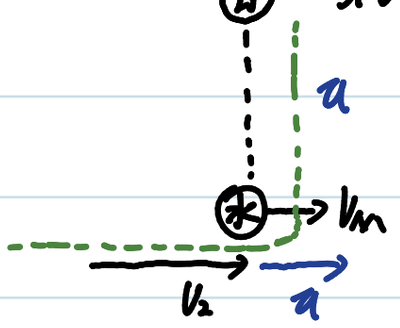
$\Rightarrow v_0^2 = \frac{2GMm}{r_E} + \frac{GM}{r_{ES}} + 2GM\left(\frac{1}{r_{ES}} - \frac{1}{r_{ES}+r_{MS}}\right)$

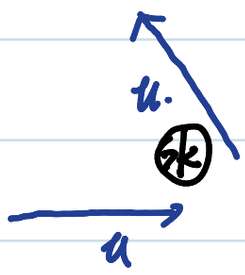
$-2GM\sqrt{\frac{1}{r_{ES}} \cdot 2\left(\frac{1}{r_{ES}} - \frac{1}{r_{ES}+r_{MS}}\right)}$.

(2) $T = 2\pi \sqrt{\frac{(r_{ES}+r_{MS})^3}{8GM}}$.

$t = \frac{1}{2} T = \pi \sqrt{\frac{(r_{ES}+r_{MS})^3}{8GM}}$

(3). 引力跳板 (以大质量为参. 避免水星速度 - 附小量级差).





条件: $v_2 - v_M > v_M$

$\Rightarrow v_2 > 2v_M$.

相对圆轨道.

$\frac{1}{2} m v_2^2 - \frac{GMm}{r_{MS}} = -\frac{GMm}{r_{ES}+r_{MS}}$

圆轨道.

$\frac{GMm}{r_{MS}^2} = m \frac{v_M^2}{r_{MS}}$.

北京质心教育科技有限公司 Center Of Mass Educational Tech. Co. Ltd

$\Rightarrow \sqrt{2GM\left(\frac{1}{r_{MS}}-\frac{1}{r_{BS}+r_{MS}}\right)} > 2\sqrt{\frac{GM}{r_{MS}}}$.

$\Rightarrow \frac{r_{BS}}{r_{BS}+r_{MS}} > 2$. 不可能

2. 流守恒.

$\oint \vec{E}\cdot d\vec{s} = q/\varepsilon_0$.

$\vec{E} = -\nabla\varphi$.

$\oint \vec{j}\cdot d\vec{s} = Q = \frac{dN}{dt}$

$\vec{j} = -D\nabla n$.

$\vec{E}$ $\quad$ φ.

打散很慢

$n(r,t) \approx n(r)$.

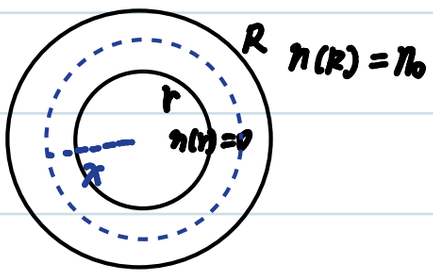
静电学.

快慢自由度问题.

漫渐不变量.

$n(r,t)$.

缓慢→参数.



流守恒: $j\cdot 4\pi r^2 = Q$.

$\Rightarrow j = \frac{Q}{4\pi r^2}$

题干: $j = D\cdot\frac{dn}{dr}$.

$\Rightarrow n = -\frac{Q}{4\pi D}\frac{1}{r} + C$.

边界 $n(r)=0$. $n(R)=n_0$.

$\Rightarrow Q = \frac{4\pi D n_0}{\frac{1}{R}-\frac{1}{r}}$

$\Rightarrow n(x) = \frac{Q}{4\pi D}\left(\frac{1}{x}-\frac{1}{r}\right)$

$\rho = \frac{n\cdot\Delta V\cdot M}{N_A\cdot\Delta V} = \frac{nM}{N_A}$.

$\Rightarrow \rho = \frac{\rho_0}{\frac{1}{R}-\frac{1}{r}}\left(\frac{1}{x}-\frac{1}{r}\right)$.

(2). $Q = \frac{\Delta N}{\Delta t} = \frac{4\pi D}{\frac{1}{R}-\frac{1}{r}}\cdot\frac{\rho_0 N_A}{M}$.

单位时摩尔数: $\frac{Q}{N_A} = \frac{4\pi D}{\frac{1}{R}-\frac{1}{r}}\frac{\rho_0}{M}$.

(3). $\frac{dQ}{dt}\cdot\frac{\frac{4}{3}\pi R^3\cdot\rho}{M_3}N_A = Q$.

(4). $Q = 1-\left(\frac{r}{R}\right)^3$

$\Rightarrow \frac{dQ}{dt} = -\frac{3r^2}{R^3}\frac{dr}{dt}$

$\Rightarrow Q = -\frac{3r^2}{R^3}\frac{dr}{dt}\frac{\frac{4}{3}\pi R^3\cdot\rho}{M_3}N_A$

$\Rightarrow 4\pi D n_0\cdot\frac{M_3\cdot R^3}{4\pi R^3\rho N_A}dt = -r^2dr\cdot\left(\frac{1}{R}-\frac{1}{r}\right)$.

$\Rightarrow \frac{D n_0 M_3}{\rho N_A}dt = -\frac{1}{R}r^2dr + r\,dr$.

$\Rightarrow \frac{D n_0 M_3}{\rho N_A}T = \frac{1}{2}r^2 - \frac{1}{3R}r^3\Big|_0^r$

3. $\vec{F} = -km_1m_2\vec{r}$

(1). $F_x = -k\mu m x$.

$F_y = -k\mu m y$.

$\Rightarrow$ 水平竖直. 简谐. $\omega = \sqrt{k\mu m}$.

初始. $x(0)=r_0$. $\dot{x}(0)=0$.

$x(t) = r_0\cos(\omega t)$.

$y(0)=0$. $\dot{y}(0)=v_0$

$\frac{1}{2}k\mu m\cdot A^2 = \frac{1}{2}mv_0^2 \Rightarrow A = \frac{v_0}{\omega} = \frac{v_0}{\sqrt{k\mu m}}$.

$y(t) = \frac{v_0}{\sqrt{k\mu m}}\sin\omega t$.

(2). $r_c = \frac{1}{3}r_0$. $y_c = \frac{1}{3}v_0 t$.

以质心为参. m 到 M 距离为 (x,y).

m 到质心距离. $r' = \frac{M}{M+m}r$.

$y' = \frac{M}{M+m}y$.

$F_x = -k\mu m r = -k\mu m\frac{m+M}{M}r'$

$F_y = -k\mu m y = -k\mu m\frac{m+M}{M}y'$.

$\Rightarrow$ 水平竖直简谐 $\omega = \sqrt{k(m+M)} = \sqrt{3k\mu}$

初始值. $r'(0) = \frac{2}{3}r_0$. $\dot{r}'(0)=0$.

$\Rightarrow r' = \frac{2}{3}r_0\cos(\omega t)$.

北京质心教育科技有限公司 Center Of Mass Educational Tech. Co. Ltd

初始: $y'(0)=0$. $\dot{y}'(0)=\frac{2}{3}v_0$.

$\Rightarrow \frac{2}{3}v_0 = \omega A$

$\Rightarrow A = \frac{2}{3}v_0\frac{1}{\sqrt{3}k\omega}$

$y'(t)=\frac{2}{3}\frac{v_0}{\sqrt{3}k\omega}\sin\omega t$.

地面系:

$x(t)=\frac{2}{3}r_0\cos\omega t+\frac{1}{3}r_0$.

$y(t)=\frac{2}{3}\frac{v_0}{\sqrt{3}k\omega}\sin\omega t+\frac{1}{3}v_0 t$.

(2). 质心系. 质心为原点..

3个解 $\vec{r}_1$. $\vec{r}_2$. $\vec{r}_3$

1受力:

$m_1\vec{a}_1 = km_3m_1(\vec{r}_3-\vec{r}_1)+km_2m_1(\vec{r}_2-\vec{r}_1)$.

$\Rightarrow \vec{a}_1 = km_3\vec{r}_3+km_2\vec{r}_2-k(m_3+m_2)\vec{r}_1$

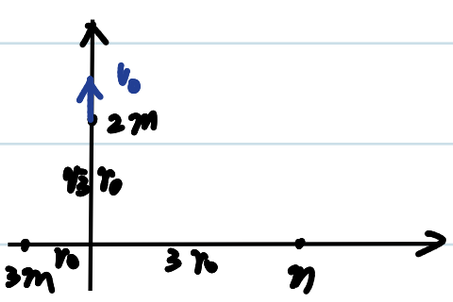
$=k(m_1\vec{r}_1+m_2\vec{r}_2+m_3\vec{r}_3)$

$-k(m_1+m_2+m_3)\vec{r}_1$

$=-6km\vec{r}_1$

即 $\omega=\sqrt{6km}$.

m. $2m$. $3m$. 在质心系里简谐.



$x_C=\frac{-3mr_0+3mr_0}{6m}=0$.

$y_C=\frac{2m\sqrt{3}r_0}{6m}+\frac{2mv_0}{6m}t$.

$=\frac{2}{3}r_0+\frac{1}{3}v_0 t$.

m. 质心系 初始

坐标 $(3r_0,-\frac{1}{\sqrt{3}}r_0)$. 速度 $(0,-\frac{1}{3}v_0)$.

$x'=3r_0\cdot\cos\omega t$

$y'=-\frac{1}{\sqrt{3}}r_0\cos\omega t-\frac{1}{3}\frac{v_0}{\omega}\sin\omega t$

地面系: $x=3r_0\cdot\cos\omega t$

$y=-\frac{1}{\sqrt{3}}r_0\cos\omega t-\frac{1}{3}\frac{v_0}{\omega}\sin\omega t+\frac{2\sqrt{3}}{3}r_0+\frac{1}{3}v_0 t$.

$2m$. 质心系. 初始. 坐标 $(0,\frac{2\sqrt{3}}{3}r_0)$. 速度 $(0,\frac{2}{3}v_0)$.

$x'=0$.

$y'=\frac{2\sqrt{3}}{3}r_0\cos\omega t+\frac{2}{3}v_0\sin\omega t$

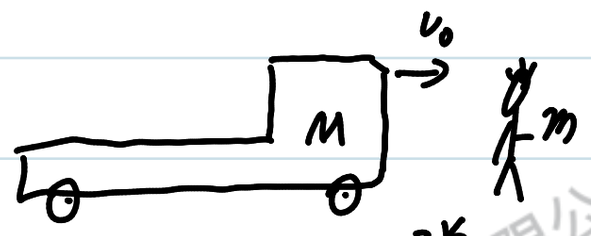
地面系:

$x=0$

$y=\frac{2\sqrt{3}}{3}r_0\cos\omega t+\frac{2}{3}v_0\sin\omega t+\frac{2\sqrt{3}}{3}r_0+\frac{1}{3}v_0 t$.

4. 20个拉格朗日点..

任置任意. 共质统质心有共用 $\omega=\sqrt{km_总}$.



初 v_0 0.

末 v_0. $2v_0$.

$\downarrow\downarrow$ $\downarrow\downarrow$

$\frac{M-m}{M+m}v_0$ $\frac{2M}{M+m}v_0$

3单.

能 初. $\frac{1}{2}Mv_0^2$.

末 $\frac{1}{2}Mv_0^2+\frac{1}{2}m(2v_0)^2$.

4. (1). $\vec{M}=I\vec{S}\times\vec{B}$

$M=Ir^2B\sin\theta\approx Ir^2B\cdot\theta$.

$T=2\pi\sqrt{\frac{J}{Ir^2B}}$.

(2). $T_0=2\pi\sqrt{\frac{J}{Ir^2B_0}}$

$T_1=2\pi\sqrt{\frac{J}{Ir^2(B_0+B_1)}}$.

$\Rightarrow \frac{T_0^2}{T_1^2}=\frac{B_0+B_1}{B_0}$

$\Rightarrow B_0=\frac{T_1^2}{T_0^2-T_1^2}B_1$.

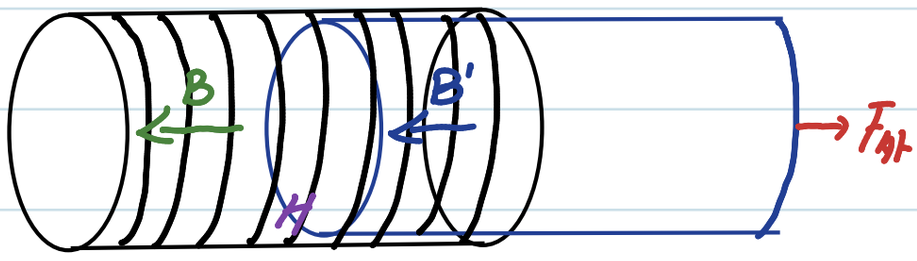
5. (1). $B\cdot l=\mu_0\cdot n\cdot l\cdot I$

$\Rightarrow B=\mu_0 nI$.

(2). $\Phi=B\cdot N\cdot S$.

$L=\frac{\Phi}{I}=\mu_0 n^2\cdot V$.

北京质心教育科技有限公司 Center Of Mass Educational Tech. Co. Ltd



$\oint \vec{H}\cdot\vec{dl} = I$. 传导电流.

$\Rightarrow H = nI$.

$B = \mu_0\mu_r H$.

不妨假设. 外界有恒定电流源. I.



能量微元.

$F_{外}dx + I\cdot d\Phi = (\frac{1}{2}\mu_0 H^2 - \frac{1}{2}\mu_0\mu_r H^2)\pi r^2 dx$.

电感: $I\,d\Phi = 2(\frac{1}{2}\mu_0 H^2 - \frac{1}{2}\mu_0\mu_r H^2)\pi r^2 dx$.

$\Rightarrow F_{外}dx = -(\frac{1}{2}\mu_0 H^2 - \frac{1}{2}\mu_0\mu_r H^2)\pi r^2 dx$.

$\Rightarrow F_{外} = \frac{1}{2}\mu_0 H^2\cdot\pi r^2(\mu_r - 1)$.

$= \frac{1}{2}\mu_0 n^2\cdot I^2\cdot\pi r^2(\mu_r - 1)$.

$F_{场} = F_{外}$. 方向向左.

(4). 回路无电阻且.

$\frac{d\Phi}{dt} + IR = 0$.

$\Rightarrow$ 更不变.

$\mu_0 nI = \mu_0\mu_r nI'$

$\Rightarrow I' = \frac{I}{\mu_r}$.

能量守恒.

$\frac{1}{2}mv^2 + \frac{1}{2}\mu_0\mu_r(nI')^2\pi r^2 l$.

$= \frac{1}{2}\mu_0(nI)^2\pi r^2\cdot l$.

$\Rightarrow v = \mu_0 nI_0 r\sqrt{\frac{\pi l}{\mu_0 m}}\sqrt{1-\frac{1}{\mu_r}}$

不变. 法向 D 连续. 切向 E 连续.

法向 B 连续. 切向 H 连续

边缘效应. $l \gg r$.



(5). (2) 问改成. 外面. 联. 3 一大堆电路.

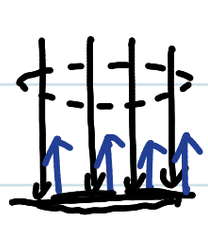
解法和答案完全用(2).

6. (1). $S = \pi r^2 = \pi\cdot(f\cdot\delta\theta)^2$.

$= \frac{\pi f^2\lambda^2}{D^2}1.22^2$

(2). 自题体有答案定理.

$\Rightarrow$ 黑体照射部分与没照部分. 没有相位差.



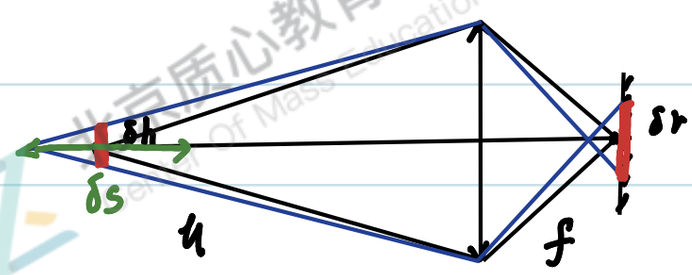


吸能 = 放能

$\frac{\Delta R}{\Delta t} = \frac{\pi D^2}{4}\sigma T_0^4$ $\frac{\Delta Q}{\Delta t} = S\cdot\sigma\cdot T'^4$.

$\Rightarrow T' = \frac{D}{\sqrt{2.44 f\lambda}}T_0$.

(3). 共轭.



$u \gg f$

$\frac{\delta r}{\delta h} = \frac{f}{u-f} \simeq \frac{f}{u}$

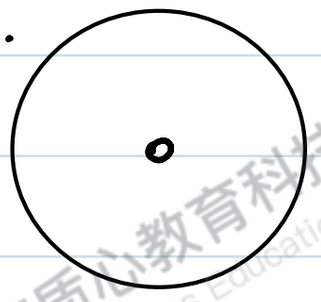
$\delta r \to \delta r + f\cdot\delta\theta$.

$\frac{\delta h}{\delta u} = \frac{\frac{D}{2}}{\delta u + u} \simeq \frac{D}{2u}$

$\Rightarrow \delta u = \frac{2u^2\delta r}{fD}$

$\Rightarrow$ 景深 $\delta s = 2\delta u = \frac{4u^2\delta r}{fD} = \frac{4\pi^2(\delta r + f\cdot\delta\theta)}{f\cdot D}$.

7. (1).



$\oint \vec{j}\cdot\vec{dS} = I$

$j = \frac{I}{4\pi r^2}$

$E = j\rho = \frac{\rho I}{4\pi r^2}$.

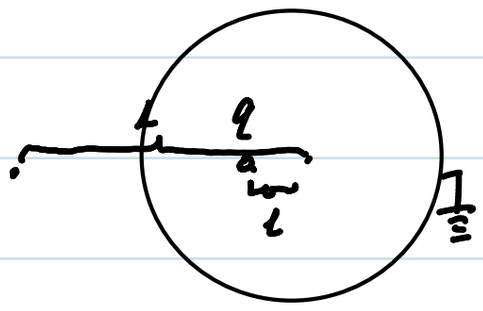
$U = \int_r^R E\cdot dx = \frac{\rho I}{4\pi}(\frac{1}{r} - \frac{1}{R})$.

$R_{组} = \frac{U}{I} = \frac{\rho}{4\pi}(\frac{1}{r} - \frac{1}{R})$.

北京质心教育科技有限公司 Center Of Mass Educational Tech. Co. Ltd

(2) $q_{内}=\oint \vec{j}\cdot d\vec{S}=\iiint \frac{\vec{j}}{\sigma}dS=\frac{I}{\sigma}$.

等效成电荷问题.



$q'=-\frac{R}{l}q$. $l=\frac{R^2}{l}$.

$U=\frac{1}{4\pi\varepsilon_0}(\frac{q}{r}+\frac{q'}{l-l})$

$=\frac{q}{4\pi\varepsilon_0}(\frac{1}{r}-\frac{R}{R^2-l^2})$.

$U=\frac{I}{4\pi\sigma}(\frac{1}{r}-\frac{R}{R^2-l^2})$.

$R_{相}=\frac{U}{I}=\frac{\rho}{4\pi}(\frac{1}{r}-\frac{R}{R^2-l^2})$.

8. (1) 地面系中. 飞船以 v 在飞行.

车行正系. 速度增加 dv'

换到地面系速度增加 dv.

$v+dv=\frac{v+dv'}{1+\frac{v\cdot dv'}{c^2}}$

地面系认为速度增加. 换系导致的速度加.

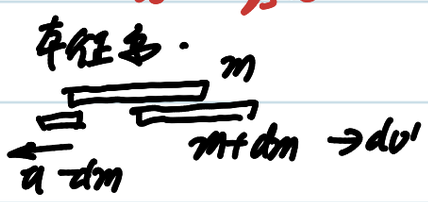
$\Rightarrow v+dv+\frac{v^2}{c^2}dv'=v+dv'$

$\Rightarrow dv=\frac{1}{\gamma^2}dv'$

加速度 $\frac{dv}{dt}=\frac{1}{\gamma^3}\frac{dv'}{dt'}$.

$a=\frac{1}{\gamma^3}a'$

车行系.



动量: $mdv'=-dm\,u$.

$\Rightarrow -\frac{dm}{m}=\frac{dv'}{u}=\frac{1}{u}\frac{dv}{1-\beta^2}$

$\Rightarrow -\frac{dm}{m}\frac{2u}{c}=(\frac{d\beta}{1-\beta}+\frac{d\beta}{1+\beta})$

$\Rightarrow \frac{M_0}{M_1}=(\frac{1+\beta}{1-\beta})^{\frac{c}{2u}}$

此题: 加速. $\frac{m_0}{m}=\sqrt{1-\beta^2}$. 减速 $\frac{m'}{m}=\sqrt{1+\beta^2}$.

$t\to 0$ 时间. 喷完. $V=\frac{S}{\gamma}$

$\Rightarrow \frac{m'}{m_0}=\frac{1-\beta}{1+\beta}$

$\Rightarrow \frac{m_{燃料}}{m_0}=1-\frac{m'}{m_0}$

(2). 飞船系中. 飞船提供的正物质静质量与反物质的静质量相同.

飞船系中: dt 时间. 飞船速度为 dv.



动: $-\gamma^2\rho S v\cdot dt\,v=(m-\gamma\rho S v dt)dv-\frac{E_{光}}{c}$.

能: $\gamma^2\rho S v\,dt\cdot c^2+mc^2=(m-\gamma\rho S v dt)c^2+E_{光}$.

$\Rightarrow \frac{dv}{dt}=\frac{\gamma\rho S v}{m}[(\gamma+1)c-\gamma v]$

换回地面系:

$\frac{dv}{dt}=\frac{1}{\gamma^3}\frac{dv}{dt}=\frac{\rho S v}{\gamma^2 m}[(\gamma+1)c-\gamma v]$

北京质心教育科技有限公司 Center Of Mass Educational Tech. Co. Ltd